

## Huaxu Yu, Ph.D.

Clinical Distinguished Researcher, The First Affiliated Hospital of Zhejiang University School of Medicine

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**The MassCube Project:** <https://huaxuyu.github.io/masscubedocs>

## Employment History

2026-present **Clinical Distinguished Researcher (临床特聘研究员)**

The First Affiliated Hospital of Zhejiang University School of Medicine

2023-2026 **Postdoctoral Researcher**

Genome Center, University of California Davis, Davis, CA, United States

Research Supervisor: Prof. Oliver Fiehn

## Education

2019-2023 **Ph.D. in Chemistry**

Department of Chemistry, The University of British Columbia, Vancouver, BC, Canada

Research Supervisor: Prof. Tao Huan

2014-2018 **B.S. in Chemistry**

Department of Chemistry, Zhejiang University, Hangzhou, Zhejiang, China

## Research Interests

- **Computational metabolomics:** breaking through limitations in accuracy and efficiency for ultra-large-scale (>10,000 samples) mass spectrometry data processing for untargeted metabolomics.
- **Cancer metabolomics:** developing AI models for representing metabolic reprogramming in tumor microenvironment to support precision oncology and clinical decision-making.

## Publication list

My research has been published in journals including *Nature Communications*, *Analytical Chemistry*, and *Nature Cancer*, with over 633 citations and an h-index of 16. As first author, I have published 9 research articles focused mass spectrometry-based untargeted metabolomics, as well as their biological and clinical applications. I also developed MassCube, an end-to-end metabolomics data processing software package with peak monthly downloads exceeding 1,200. A full list of my publications is provided below.

- 1 **Yu, H.**, Ding, J., Shen, T., Liu, M., Li, Y., Fiehn, O. MassCube: a Python framework for end-to-end metabolomics data processing from raw files to phenotype classifiers.  
*Nature Communications*, **2025**, 16(1), 5487. <https://doi.org/10.1038/s41467-025-60640-5>  
**Project website:** <https://huaxuyu.github.io/masscubedocs>  
**Software download:** [83.56k](#)
- 2 **Yu, H.**, Low, B., Zhang, Z., Guo, J., Huan, T. Quantitative challenges and their bioinformatic solutions in mass spectrometry-based metabolomics. (review article)  
*Trends in Analytical Chemistry*, **2023**, 161, 117009. <https://doi.org/10.1016/j.trac.2023.117009>
- 3 **Yu, H.**, Huan, T. MAFFIN: metabolomics sample normalization using maximal density fold change with high-quality metabolic features and corrected signal intensities.  
*Bioinformatics*, **2022**, 38(13), 3429-3437. <https://doi.org/10.1093/bioinformatics/btac355>
- 4 **Yu, H.**, Sang, P., Huan, T. Adaptive Box-Cox transformation: a highly flexible feature-specific data transformation to improve metabolomics data normality for better statistical analysis.  
*Analytical Chemistry*, **2022**, 94(23), 8267–8276. <https://doi.org/10.1021/acs.analchem.2c00503>
- 5 **Yu, H.**, Huan, T. Comprehensive assessment of the diminished statistical power caused by nonlinear electrospray ionization responses in mass spectrometry-based metabolomics.  
*Analytica Chimica Acta*, **2022**, 1200, 339614. <https://doi.org/10.1016/j.aca.2022.339614>
- 6 **Yu, H.**, Huan, T. Patterned signal ratio biases in mass spectrometry-based quantitative metabolomics.  
*Analytical Chemistry*, **2021**, 93(4), 2254–2262. <https://doi.org/10.1021/acs.analchem.0c04113>
- 7 **Yu, H.**, Chen, Y., Huan, T. Computational variation: An under-investigated quantitative variability caused by automated data processing in untargeted metabolomics.  
*Analytical Chemistry*, **2021**, 93(25), 8719–8728. <https://doi.org/10.1021/acs.analchem.0c03381>
- 8 **Yu, H.**, ..., Huan, T. Fold-Change compression: An unexplored but correctable quantitative bias caused by nonlinear electrospray ionization responses in untargeted metabolomics.  
*Analytical Chemistry*, **2020**, 92(10), 7011–7019. <https://doi.org/10.1021/acs.analchem.0c00246>
- 9 **Yu, H.**, Xing, S., Nierves, L., Lange, PF., Huan, T. Parallel metabolomics and lipidomics

enables the comprehensive study of mouse brain regional metabolite and lipid patterns.

*Analytica Chimica Acta*, **2020**, 1136, 168-177. <https://doi.org/10.1016/j.aca.2020.09.051>

- 10 Biswas, P., Bako, J. A., Liston, J. B., **Yu, H.**, ..., Rideout, E. J. Insulin/insulin-like growth factor signaling pathway promotes higher fat storage in *Drosophila* females.  
*Cell Reports*, **2025**, 44(7). <https://doi.org/10.1016/j.celrep.2025.115915>
- 11 Zhao, L., Qiu, Z., Yang, Z., Xu, L., Pearce, T.M., Wu, Q., Yang, K., Li, F., Saulnier, O., Fei, F., **Yu, H.**, ..., Rich, J. Lymphatic endothelial-like cells promote glioblastoma stem cell growth through cytokine-driven cholesterol metabolism.  
*Nature Cancer*, **2024**, 5(1), 147-166. <https://doi.org/10.1038/s43018-023-00658-0>
- 12 Low, B., Wang, Y., Zhao, T., **Yu, H.**, Huan, T. Closing the Knowledge Gap of Post-Acquisition Sample Normalization in Untargeted Metabolomics.  
*ACS Meas. Sci. Au*, **2024**, 4(6), 702-711. <https://doi.org/10.1021/acsmeasuresciau.4c00047>
- 13 Chen, Y., Wang, Y., Delgado, D. H., **Yu, H.**, Zhao, T., Fang, M., Huan, T. Constructing HairDB to facilitate exposome research using human hair.  
*Environment International*, **2024**, 193, 109077. <https://doi.org/10.1016/j.envint.2024.109077>
- 14 Zhang, Z., **Yu, H.**, ..., Huan, T. Reducing Quantitative Uncertainty Caused by Data Processing in Untargeted Metabolomics.  
*Analytical Chemistry*, **2024**, 96(9), 3727-3732. <https://doi.org/10.1021/acs.analchem.3c04046>
- 15 Zhao, T., Xing, S., **Yu, H.**, Huan, Tao. De novo cleaning of chimeric MS/MS spectra for LC-MS/MS-based metabolomics.  
*Analytical Chemistry*, **2024**, 95(35), 13018-13028. <https://doi.org/10.1021/acs.analchem.3c00736>
- 16 Zhao, T., Wawryk, N.J., Xing, S., Low, B., Li, G., **Yu, H.**, ..., Huan, T. ChloroDBPFinder: machine learning-guided recognition of chlorinated disinfection byproducts from nontargeted LC-HRMS analysis.  
*Analytical Chemistry*, **2024**, 96(6), 2590-2598. <https://doi.org/10.1021/acs.analchem.3c05124>
- 17 Jandu, R. S., **Yu, H.**, Zhao, Z., Le, H. T., Kim, S., Huan, T., van Hoa, F. D., Capture of endogenous lipids in peptidiscs and effect on protein stability and activity.  
*iScience*, **2024**, 27(4), 109382. <https://doi.org/10.1016/j.isci.2024.109382>
- 18 Chao, C. F., Pesch, Y. Y., **Yu, H.**, ..., Rideout, E. An important role for triglyceride in regulating spermatogenesis.  
*eLife*, **2024**, 12, RP87523. <https://doi.org/10.7554/eLife.87523.4>
- 19 Guo, J., Shen, S., Liu, M., Wang, C., Low, B., Chen, Y., Hu, Y., Xing, S., **Yu, H.**, Gao, Y., Fang, M., Huan, T. JPA: Joint metabolic feature extraction increases the depth of chemical coverage for LC-MS-based metabolomics and exposomics.

- Metabolites*, **2022**, 12(3), 212. <https://doi.org/10.3390/metabo12030212>
- 20 Guo, J., **Yu, H.**, Xing, S., Huan, T. Addressing big data challenges in mass spectrometry-based metabolomics.  
*Chemical Communications*, **2022**, 58(72), 9979-9990. <https://doi.org/10.1039/D2CC03598G>
  - 21 Xing, S., **Yu, H.**, Liu, M., Jian, Q., Sun, Z., Fang, M., Huan, T. Recognizing contamination fragment ions in liquid chromatography-tandem mass spectrometry data.  
*J. Am. Soc. Mass Spectrom.*, **2021**, 32(9), 2296-2305. <https://doi.org/10.1021/jasms.0c00478>
  - 22 Chen, Y., Guo, J., Xing, S., **Yu, H.**, Huan, T. Global-scale metabolomic profiling of Human Hair for simultaneous monitoring of endogenous metabolome, short-and long-term exposome.  
*Frontiers in chemistry*, **2021**, 9, 674265. <https://doi.org/10.3389/fchem.2021.674265>
  - 23 Guo, J., Shen, S., Xing, S., **Yu, H.**, Huan, T. ISFrag: De novo recognition of in-source fragments for liquid chromatography–mass spectrometry data.  
*Analytical Chemistry*, **2021**, 93(29), 10243-10250. <https://doi.org/10.1021/acs.analchem.1c01644>
  - 24 Guo, J., Shen, S., Xing, S., Chen, Y., Chen, F., Porter, E.M., **Yu, H.**, Huan, T. EVA: Evaluation of Metabolic Feature Fidelity Using a Deep Learning Model Trained with Over 25000 Extracted Ion Chromatograms.  
*Analytical Chemistry*, **2021**, 93(36), 12181-12186. <https://doi.org/10.1021/acs.analchem.1c01309>
  - 25 Sun, Y., Yao, Y., Wang, H., Fu, W., Chen, C., Saha, M. L., Zhang, M., Datta, S., Zhou, Z., **Yu, H.**, Li, X., Stang, P. J. Self-assembly of metallacages into multidimensional suprastructures with tunable emissions.  
*J. Am. Chem. Soc.*, **2018**, 140(40), 12819-12828. <https://doi.org/10.1021/jacs.8b05809>
  - 26 Yao, Y., Sun, Y., **Yu, H.**, Chen, W., Dai, H., Shi, Y. A pillar[5]arene based gel from a low-molecular-weight gelator for sustained dye release in water.  
*Dalton Transactions*, **2017**, 46(48), 16802-16806. <https://doi.org/10.1039/C7DT04001F>

#### Publications (non-refereed)

- 27 **Yu, H.**, ..., Fiehn, O. Inferring dietary exposure using cross-over mixed meals and discovery metabolomics.  
*Proceedings of the National Academy of Sciences*. **2026**. In revision.
- 28 **Yu, H.**, Biswas, P., Rideout, E., Cao, Y., Huan, T. Bayesian optimization of separation gradients to maximize the performance of untargeted LC-MS.  
*Analytical Chemistry*. **2026**. In revision. <https://doi.org/10.1101/2023.09.08.556930>
- 29 Liu, M., Ji, J., David, A., **Yu, H.**, ..., Fiehn O. The Metabolome Atlas of 22 Tissues in Aging Mice Reveals a Switch in Thermogenesis from Brown Fat to Skeletal Muscle.

*Nature Metabolism*, 2025. In review. <https://www.researchsquare.com/article/rs-7811947/v1>

- 30 Zhu, J., Liu, G., Xu, J., Li, K., Goins, C. M., **Yu, H.**, ..., Gack, M. U. Dysregulation of innate immunity and cellular metabolism through virus-induced deISGylation.

*bioRxiv*, 2025. <https://www.biorxiv.org/content/10.1101/2025.09.08.674928v1.abstract>

- 31 Xing, S., Charron-Lamoureux, V., Ekelöf, M., El Abiead, Y., **Yu, H.**, ..., Dorrestein, P. C. Structural annotation of full-scan MS data: A unified solution for LC-MS and MS imaging analyses.

*Nature Communications*, 2025. In revision.

<https://www.biorxiv.org/content/10.1101/2024.10.14.618269v2>

## Talks

- Jun. 2026 DIET-IF: dietary exposure inference from mixed meals using cross-over metabolomics. *74<sup>th</sup> ASMS Conference on Mass Spectrometry and Allied Topics*. San Diego Convention Center, San Diego, California, USA
- Jan. 2026 VMOL Seminar #45 - Mass Spectrometry Feature Detection with MassCube. *The VMOL Seminar Series. Invited by Prof. Daniel Petras*. Recording: <https://www.youtube.com/watch?v=ymzwh3f-47Q>.
- Jun. 2025 MassCube: a Python framework for end-to-end metabolomics data processing from raw files to phenotype classifiers. *73<sup>rd</sup> ASMS Conference on Mass Spectrometry and Allied Topics*. Baltimore Convention Center, Baltimore, Maryland, USA
- May 2025 Uncovering dietary biomarkers through human postprandial metabolic responses. *West Coast Metabolomics Center Seminar*. University of California, Davis, Davis, USA
- Apr. 2025 MassCube: a Python framework for end-to-end metabolomics data processing from raw files to phenotype classifiers. *The Software Data Exchange (SODA) Meetup*. Recording: <https://www.youtube.com/watch?v=oxQAnanqbT8>.
- Oct 2024 Analytical and bioinformatic solutions for mass spectrometry-based untargeted metabolomics. *Invited talk by Prof. Feng Zhu*, College of Pharmaceutical Sciences, Zhejiang University, Hangzhou, China.
- Jan 2024 Confident structural identification of small molecule using retention time standardization. *West Coast Metabolomics Center Seminar*. University of California, Davis, Davis, USA
- Oct 2023 Data alignment of untargeted LC-MS/MS experiments using internal standards. *37<sup>th</sup> Asilomar Conference on Mass Spectrometry Security and Forensic*

## Curriculum Vitae

*Applications of Mass Spectrometry*, Asilomar Conference Center, Asilomar, CA, USA

- Oct 2023 Accurate annotation of metabolite mass spectra using deep learning. *Virtual Metabolomics Journal Club*. Online meeting.
- Jun 2023 Development of analytical and bioinformatic solutions for quantitative metabolomics. *West Coast Metabolomics Center Seminar*. University of California, Davis, Davis, USA
- Mar 2023 Development of analytical workflows and bioinformatic programs for mass spectrometry-based metabolomics. *Ph.D. defense*. Department of Chemistry, University of British Columbia, Canada
- Jan 2023 Integrated method development of quantitative metabolomics using conventional analytical chemistry and machine learning. *Invited talk by Prof. Joshua Rabinowitz*. Princeton University, Princeton, NJ, USA
- Oct 2022 MAFFIN: metabolomics sample normalization using maximal density fold change with high-quality metabolic features and corrected signal intensities. *The 2<sup>nd</sup> CASMS Virtual Conference*. Online meeting.
- Oct 2021 Patterned signal ratio biases in mass spectrometry-based quantitative metabolomics. *The 1<sup>st</sup> CASMS Virtual Conference*. Online meeting.

## Posters

- Jun 2024 Standardizing retention times to reduce ambiguity of small molecule identification. *72<sup>nd</sup> ASMS Conference on Mass Spectrometry and Allied Topics*. Anaheim Convention Center, Anaheim, CA, USA
- Jun 2023 Sexual dimorphism of rewarding system in mouse brain revealed by parallel metabolomics and lipidomics. *71<sup>st</sup> ASMS Conference on Mass Spectrometry and Allied Topics*. George R. Brown Convention Center, Houston, TX, USA
- Jun 2022 Fold change bases in mass spectrometry-based quantitative metabolomics: causes and solutions. *70<sup>th</sup> ASMS Conference on Mass Spectrometry and Allied Topics*.
- Jun 2020 Calibrating nonlinear ESI responses using quality control samples to overcome quantitative errors in mass spectrometry-based metabolomics. *68<sup>th</sup> ASMS Conference on Mass Spectrometry and Allied Topics*. Online meeting.

## Expertise

Mass spectrometry (MS)

Extensive hands-on experience with Bruker Impact II Q-TOF MS, ThermoFisher Q Exactive Orbitrap MS, Orbitrap Exploris™ 240 MS, TSQ Plus Triple Quadrupole MS, Bruker timsTOF Pro 2 MS and SCIEX TripleTOF 6600+ MS.

### Software development

Sole developer and maintainer of [MassCube](#) Python package, [BAGO](#) Python package, [BAGO](#) Windows software, [ABCstats](#) R package and [MAFFIN](#) R package.

### Machine learning and artificial intelligence

Gaussian processing regression-guided Bayesian optimization workflow for liquid chromatography configuration design (developed the BAGO software). Deep learning-based chromatographic peak recognition (contributed to the EVA software). Phenotype classifier tool based on random forest model (developed and implemented in MassCube).

## **Teaching**

- |         |                                                                                                                                                                         |
|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2024. 8 | Introduction: Demo on MassCube software. International Sessions in Metabolomics and Exosome Studies 2024                                                                |
| 2024. 8 | Quantification in untargeted analysis: using serial dilutions. International Sessions in Metabolomics and Exosome Studies 2024                                          |
| 2024. 8 | Quantification in targeted analyses: using internal standards, NIST reference materials, MRMs and kits. International Sessions in Metabolomics and Exosome Studies 2024 |
| 2024. 8 | Data processing on Compound Discoverer, MassCube, Skyline. International Sessions in Metabolomics and Exosome Studies 2024                                              |
| 2024. 8 | mz-rt peak grouping in MS-based untargeted small molecule analysis. International Sessions in Metabolomics and Exosome Studies 2024                                     |
| 2024. 5 | Quantification in Metabolomics: Tools for Robustness. WCMC Bits & Bites #4, 2024                                                                                        |
| 2023. 8 | Recent advances of discovering feature relations in MS-based untargeted small molecule analysis. WCMC metabolomics summer course 2023                                   |
| 2023. 8 | Improving quantitative accuracy in untargeted small molecule analysis using serial diluted QC samples and computational tools. WCMC metabolomics summer course 2023     |

## Awards

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|---------|-------------------------------------------------------------------------------------------------------------------|
| 2023.8  | ASMS Asilomar Conference Travel Award, American Society for Mass Spectrometry                                     |
| 2022.9  | President's Academic Excellence Initiative PhD Award, The University of British Columbia                          |
| 2021.9  | Pei-Huang Tung and Tan-Wen Tung Graduate Fellowship, The University of British Columbia                           |
| 2021.9  | Gladys Estella Laird Research Fellowship, The University of British Columbia                                      |
| 2021.9  | President's Academic Excellence Initiative PhD Award, The University of British Columbia                          |
| 2019.5  | Chemistry Graduate Fellowship, The University of British Columbia                                                 |
| 2018.6  | Award of Graduation with Distinction of Zhejiang Province (Provincial level, <b>top 1%</b> ), Zhejiang University |
| 2017.12 | First-Class Scholarship for Distinguished Students in Basic Science (top 5%), Zhejiang University                 |
| 2017.12 | Outstanding Undergraduate, Department of Chemistry, Zhejiang University                                           |
| 2016.12 | First-Class Scholarship for Distinguished Students in Basic Science (top 5%), Zhejiang University                 |
| 2016.12 | Outstanding Undergraduate, Department of Chemistry, Zhejiang University                                           |
| 2015.12 | First-Class Scholarship for Distinguished Students in Basic Science (top 5%), Zhejiang University                 |
| 2015.12 | Scholarship for Distinguished Students in Chemistry, Zhejiang University                                          |